SG-END-001 Diabetes Guideline: International Comparison and Value for Endocrinologists

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Summary

SG-END-001 Rev 1.2 is an educational evidence synthesis that presents diabetes and its subtypes in 17 sections, based on 152 numbered references, in Turkish, English, Albanian and German within a single file. More authoritative resources exist internationally, but each covers only part of the field and most are available in English only.

The guideline's distinguishing feature is that it brings together clinical management, subtype differentiation, genetic testing and variant interpretation, acute-crisis protocols and a rule-based educational simulation in one place, usable offline. Its greatest value for endocrinologists is that it combines, in a single workflow, the steps of recognising missed monogenic diabetes, ordering the right test, reading the result against ACMG/AMP and ClinGen criteria, and choosing genotype-guided treatment. The guideline does not replace a clinical practice guideline; it should be used as a reference alongside international guidelines.

1. Similar international work

Short answer: The search did not identify any resource that combines all of SG-END-001's components (all diabetes subtypes; variant interpretation and reporting in monogenic diabetes; an educational decision-support module with rule–evidence traceability; downloadable tables) **in a single document and in four languages (TR/EN/SQ/DE)**. Each individual component, however, has strong and well-established international counterparts; SG-END-001 does not replace them but brings them together in one place in a traceable way. This was not a systematic literature review; the table below relies only on sources that were opened and verified or that appear in the guideline's reference list.

Resource	Type and scope	Language	Genetic/monogenic depth	How SG-END-001 differs
ADA Standards of Care in Diabetes—2026 (Diabetes Care 49, Suppl 1)	Comprehensive clinical practice standard updated annually; all diabetes types	English	Monogenic diabetes and indications for genetic testing in the diagnosis and classification section	Builds on ADA 2026 and adds variant-reporting fields, a cascade-testing workflow and four languages
ISPAD 2022 – Monogenic diabetes chapter (Greeley et al., 2022)	Consensus guideline on diagnosis and management of monogenic diabetes in children and adolescents	English	High (MODY, neonatal diabetes, syndromic forms)	Combines paediatric content with adult endocrinology, pregnancy and kidney modules in the same document
ADA/EASD Precision Medicine Consensus Report (Chung et al., 2020) and PMDI Second International Consensus Report (Tobias et al., 2023)	Evidence appraisal of precision diabetes medicine and gaps in clinical translation	English	Conceptual framework; presents monogenic diabetes as the example of precision diagnosis that is “feasible today”	Turns this framework into clinic-ready workflows, tables and checklists
University of Exeter – diabetesgenes.org	Genetic testing service and clinician resources: 25-gene NGS panel, MODY probability calculator, VUS interpretation service, monogenic clinic finder	English	Very high (laboratory and service level)	A clinical guideline rather than a service site; points to the Exeter calculator and explains how to translate the result into clinical management
ClinGen Monogenic Diabetes Expert Panel – HNF1A ACMG/AMP specifications v3.0.0 (2025)	Gene-specific variant classification rules	English	Very high (laboratory/curator level)	Simplifies for clinicians: reading reports, managing VUS, reclassification and cascade-testing decisions

Resource	Type and scope	Language	Genetic/monogenic depth	How SG-END-001 differs
Endotext – "Diagnosis and Clinical Management of Monogenic Diabetes" (Gandolfi, Philipson, Naylor; last updated 12 August 2026)	Online textbook chapter	English	High	Offers decision points, four languages, downloadable tables and a decision-support simulation instead of textbook narrative
Albanian Ministry of Health – Primary-care Clinical Guideline for Type 2 Diabetes (approved 18 March 2022)	Primary-care management of T2D	Albanian	Monogenic diabetes is only named; no guidance on genetic testing	One of the few resources offering genetic diagnosis, subtype differentiation and specialist-referral content in Albanian (SQ)
DDG Praxisempfehlungen 2025 (German Diabetes Society)	Separate practice recommendations by topic (e.g. treatment of type 2 diabetes)	German	Genetic content not separately assessed in this review	Presents German content simultaneously with three other languages on the same evidence base
TEMD diabetes guidelines and TEMD Diabetes Technologies Guideline 2025	National clinical guidelines	Turkish	Genetic content not separately assessed in this review; the current edition year of the TEMD diagnosis–treatment guideline could not be confirmed [unverified]	Does not replace the national guideline; complements international genetic resources in Turkish in a practice-oriented way

No Kosovo-specific diabetes clinical guideline was found in this search.

Assessment: Internationally, the "best" resources are fragmented: the clinical standard (ADA), paediatric consensus (ISPAD), variant classification (ClinGen), testing service and calculator (Exeter) and precision-medicine framework (ADA/EASD, PMDI) sit in separate places and are largely English-only. SG-END-001's original contribution is not to generate new evidence but to unite these scattered resources into a single, traceable, multilingual clinical reference document.

2. General benefits for physicians

A single, verifiable point of reference. In 17 sections the guideline covers all diabetes subtypes, from epidemiology and classification to diagnostic pathways and pharmacology, surgery and β-cell replacement, and reproductive and psychosocial effects. All 152 references are given with a DOI, PMID or official link; information that could not be verified was either excluded or explicitly flagged. A physician can trace the basis of any recommendation back to its primary source in a few clicks.

A current evidence base. Content is built on the ADA Standards of Care 2026, the KDIGO 2022 guideline on diabetes in chronic kidney disease, and recent major cardiovascular and kidney outcome trials (e.g. FLOW, SOUL, SURPASS-CVOT). The 29 trials in the clinical-studies section can be compared in a single table with sample size, effect size and confidence intervals.

The same content in four languages. The Turkish, English, Albanian and German versions share the same evidence base and terminology. This reduces terminology drift in multinational teams, in cross-border referral and consultation correspondence, and when physicians trained in different languages work together. For Albanian, no separate practice-oriented resource on monogenic diabetes and genetic testing was found in this search.

Reusable tables and figures. All 61 tables can be downloaded as CSV or HTML and transferred directly into in-house protocols, teaching presentations, quality projects and research data-collection forms. The 20 figures are vector graphics drawn from scratch for the guideline; they contain no images taken from external sources and therefore carry no copyright uncertainty.

A structure that makes evidence certainty visible. GRADE evidence profiles and Evidence-to-Decision (EtD) tables are provided for SGLT2 inhibitors and teplizumab; harm data (e.g. rates of lymphopenia, rash and cytokine release syndrome in TN-10) are shown side by side with benefit data. Self-assessment tables against AGREE II, RIGHT and international reporting criteria show the physician the guideline's strengths and weaknesses openly.

Safety-focused practical content. Hypoglycaemia management, sick-day rules, crisis alerts for diabetic ketoacidosis and the hyperosmolar state, eGFR-based drug selection and label-based safety thresholds for teplizumab are presented in a form usable directly in daily practice.

An educational decision-support module. The module simulates the steps of diagnostic category, acute crisis, subtype, glycaemic target and treatment pathway, linking every rule to its evidence source (rule–evidence traceability). In resident training and case

discussions it allows the question “why was this decision suggested?” to be answered with a source. The module does not replace clinical judgement and has not undergone prospective clinical validation (see Section 5).

3. Value for endocrinologists in knowledge and genetics

3.1 Why it matters: misclassified diabetes

Monogenic diabetes is frequently misclassified as type 1 or type 2 diabetes (Greeley et al., 2022). Data from the United Kingdom show both the scale of the problem and a way to address it: in 2009 more than 80% of UK cases were estimated to be misdiagnosed; over 10 years the number of genetically confirmed monogenic diabetes cases rose more than threefold to 3,860, and estimated prevalence rose to 248 per million. The authors attributed this increase to targeted next-generation sequencing, education delivered through Genetic Diabetes Nurses, and the MODY probability calculator. The positive test rate was 33% in referrals that used the calculator versus 25% in those that did not, and 32% in referrals with Genetic Diabetes Nurse involvement versus 23% without (Pang et al., 2022). On the paediatric side, more than 10% of antibody-negative children in one national registry were reported to have monogenic diabetes (Gandolfi, Philipson and Naylor, Endotext, 2026).

These findings underpin the core value SG-END-001 offers endocrinologists: structured case-finding, use of a probability calculator and clinician education are the components shown to raise diagnostic yield, and the guideline brings these components together into a single clinical workflow. The guideline's own effect on diagnostic rates has not yet been measured.

3.2 The guideline's concrete genetic tools

Clinical question	Tool in the guideline	Value for the endocrinologist	Evidence basis
Whom should I test?	Four-step testing recommendation: urgent neonatal panel for every infant diagnosed before 6 months; MODY calculator and targeted NGS panel in autoantibody-negative individuals with preserved C-peptide, onset <35 years and a family history; m.3243A>G, WFS1, PPARG/LMNA/INSR when syndromic clues are present	Standardises testing indications; reduces unnecessary and missed test requests	Greeley 2022 (ISPAD); Shields 2012 (MODY calculator); ADA 2026
Which genes must the panel include at minimum?	At least GCK, HNF1A, HNF4A, HNF1B, KCNJ11, ABCC8, INS; including copy-number variants (CNV)	Serves as a checklist when choosing a laboratory and judging report coverage	ISPAD 2022
How should I read the variant report?	Reporting table covering ACMG/AMP classification, ClinGen HNF1A gene-specific specifications (version 3.0.0), HGVS c./p. notation, CNV, methylation and mtDNA heteroplasmy fields	Makes it easier to spot missing report fields and ask the laboratory the right question	Richards 2015; ClinGen MDEP 2025
What should I do with a VUS result?	A VUS is not used for treatment decisions; when a variant is reclassified, the patient and relatives are informed again	Prevents inappropriate treatment changes; defines follow-up responsibility	Richards 2015; ClinGen 2025

Clinical question	Tool in the guideline	Value for the endocrinologist	Evidence basis
Does a genetic diagnosis change treatment?	Genotype-guided treatment table: no treatment needed in GCK (in pregnancy, decision by fetal genotype); low-dose sulfonylurea in HNF1A/HNF4A; switch from insulin to high-dose sulfonylurea in KCNJ11/ABCC8 neonatal diabetes; in m.3243A>G avoid metformin, start insulin early, and monitor hearing and the heart	Opportunity to remove unnecessary insulin or drug burden and avoid inappropriate drugs	Pearson 2003; Pearson 2006; ISPAD 2022
What should I tell the family?	Cascade-testing workflow and genetic-counselling section; 50% risk counselling for first-degree relatives in autosomal dominant forms; recording with a pedigree	Extends the diagnosis from one patient to the family; one diagnosis benefits several people	ISPAD 2022
How useful is genetics in type 1 and type 2 diabetes?	States clearly that HLA and GWAS loci are not individual diagnostic tools, and that a type 1 genetic risk score may help in differential diagnosis	Prevents over-interpretation of polygenic tests	Genetics section of the guideline and its references

3.3 Non-genetic knowledge areas critical to endocrinology practice

Type 1 diabetes staging and disease-modifying therapy. Definitions of stages 1–3 (Insel et al., 2015), stage-specific monitoring and clinical-trial options, benefit and harm data from the TN-10 trial of teplizumab in stage 2 (Herold et al., 2019), and FDA label-based safety thresholds, vaccination and monitoring rules are presented together. As autoantibody screening becomes more widespread, this is becoming a new area of responsibility for endocrinologists.

Intermediate and rare forms. A C-peptide-based treatment algorithm for LADA (Buzzetti et al., 2020), pancreatogenic (type 3c) diabetes, cystic fibrosis-related diabetes, post-transplant and drug-induced diabetes, mitochondrial diabetes and syndromic forms are included in the differential-diagnosis tables. The guideline stresses that subtype assignment may be uncertain at diagnosis and must be reassessed using C-peptide, autoantibodies, family history and clinical course.

Cardio-renal-metabolic treatment decisions. Decisions on SGLT2 inhibitors, metformin and GLP-1 receptor agonists by eGFR category are given with KDIGO 2022 recommendation numbers and evidence grades; recent outcome trials such as FLOW, SOUL and SURPASS-CVOT appear in a comparative table.

3.4 How much benefit can be expected?

There is as yet no usage or outcome study showing the guideline's effect on endocrinologists; therefore no numerical benefit is claimed. The expected contribution can be defined on an evidence basis as follows: (1) bringing tools shown to raise diagnostic yield in monogenic diabetes case-finding (calculator, targeted panel, clinician education) into the outpatient clinic; (2) closing knowledge gaps on genotype-guided treatment, reading variant reports and cascade testing; (3) easier communication within teams and across borders through consistent terminology in four languages. Measuring this contribution requires the validation steps proposed in Section 5.

4. Use cases

Suspected MODY in the outpatient clinic. For a patient diagnosed at age 24, autoantibody-negative, with preserved C-peptide and a parent with early-onset diabetes, the physician follows the guideline's testing recommendation: estimates probability with the Exeter MODY calculator, selects a laboratory using the minimum gene list and CNV coverage, and checks the fields in the reporting table when the result arrives. For a pathogenic HNF1A variant, the genotype-guided treatment table and cascade-testing workflow are used; for a VUS, treatment is not changed and follow-up for reclassification is planned.

Neonatal and paediatric endocrinology. Ordering urgent genetic testing in an infant diagnosed before 6 months without waiting for autoantibody results, and the option of switching to sulfonylurea in a KATP channel variant, are stated explicitly. Rules for fluids and monitoring in paediatric diabetic ketoacidosis are in the same document.

Pregnancy and preconception counselling. Information such as basing treatment decisions on fetal genotype in a pregnant woman carrying a GCK variant, and the risk of macrosomia and neonatal hypoglycaemia in HNF4A, is presented alongside content on gestational and pregestational diabetes.

A family member at risk of type 1 diabetes. For an autoantibody-positive sibling, staging, eligibility for teplizumab in stage 2, laboratory safety thresholds and the harm profile are checked against the guideline; the decision is made through shared decision-making and according to national licensing.

An adult who develops an early need for insulin. When LADA is suspected, the C-peptide-based treatment algorithm is used; with hearing loss and a maternal-inheritance history, the mitochondrial diabetes section and the warning to avoid metformin apply.

Education, quality and research. In resident training, the decision-support simulation can be used for case discussion with rule–evidence mapping. The downloadable tables can form the basis of in-house protocols, quality-indicator tracking and research data-collection forms. The four-language structure provides a shared set of clinical definitions for multicentre and multinational projects.

5. Limitations and positioning

To convey the guideline's value accurately, its limits must also be stated clearly:

- **It is not an official clinical practice guideline.** SG-END-001 is an evidence synthesis prepared for education and reference; it does not replace ADA, ISPAD, KDIGO or national guidelines (e.g. TEMD). National licensing, reimbursement and laboratory access differ between countries.
- **There is no named multidisciplinary panel and no formal GRADE process.** The GRADE and Evidence-to-Decision tables in the guideline are preliminary assessments by the author team; individual conflict-of-interest declaration forms and external peer review are not part of this version. AGREE II scores are self-assessments, not ratings by independent appraisers.
- **The decision-support module has not been validated.** The module is a rule-based educational simulation; no prospective clinical validation, usability study or medical-device assessment has been performed. If it moves into clinical use, the relevant regulatory frameworks (e.g. the EU Medical Device Regulation, the EU AI Act, IEC 82304-1) must be assessed separately.
- **Its effectiveness has not been measured.** The guideline's effect on physician knowledge, test-ordering quality or patient outcomes has not yet been studied.
- **There is no separate patient version.** The guideline is aimed at physicians; for patient education it only points to a limited number of patient-information resources.

Positioning. Within these limits, SG-END-001 is strongest as a traceable, multilingual **reference and educational tool** that translates international genetic and clinical resources into the endocrinologist's day-to-day decision points.

Proposed steps to demonstrate effectiveness:

1. AGREE II scoring by independent appraisers and formation of a named, multidisciplinary expert panel.
2. Measurement of knowledge among endocrinology residents and specialists using a pre-test/post-test design (e.g. MODY indications, reading variant reports, genotype-guided treatment).
3. Comparison of the number of genetic test requests, the positive test rate and the cascade-testing rate before and after use of the guideline.
4. Retrospective concordance analysis using real case files and usability testing for the decision-support module.

6. Conclusion

High-quality international resources exist for the clinical management of diabetes, monogenic diabetes and precision diabetes medicine; however, they are published by different organisations, in different formats and largely in English only. The search did not identify any resource that combines all diabetes subtypes, variant interpretation and reporting, cascade testing, genotype-guided treatment and a decision-support simulation with rule–evidence traceability in a single document and in four languages.

For the endocrinologist, the guideline's most concrete contribution lies in genetics: it gives sourced, practice-oriented answers to the questions of whom to test, which genes a panel must include, how to read the report, how to manage a VUS, how a genetic result changes treatment and how it should be communicated to the family. The UK experience shows that widespread use of the same components (probability calculator, targeted panel, clinician education) can substantially increase monogenic diabetes diagnoses. Whether SG-END-001 achieves this effect among its own users must be measured through the studies proposed in Section 5.

This assessment is provided for information only; clinical decisions should always take into account current national and international guidelines and the individual patient's circumstances.

References

International comparison and genetics

1. American Diabetes Association Professional Practice Committee. 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes—2026. *Diabetes Care*. 2026;49(Suppl 1):S27–S49. doi:10.2337/dc26-S002

2. Greeley SAW, et al. ISPAD Clinical Practice Consensus Guidelines 2022: The diagnosis and management of monogenic diabetes in children and adolescents. *Pediatr Diabetes*. 2022;23(8):1188–1211. doi:10.1111/pedi.13426
3. Chung WK, Erion K, Florez JC, et al. Precision medicine in diabetes: a Consensus Report from the ADA and the EASD. *Diabetologia*. 2020;63(9):1671–1693. doi:10.1007/s00125-020-05181-w
4. Tobias DK, et al. Second international consensus report on gaps and opportunities for the clinical translation of precision diabetes medicine. *Nat Med*. 2023;29(10):2438–2457. doi:10.1038/s41591-023-02502-5
5. Pang L, Colclough KC, Shepherd MH, et al. Improvements in awareness and testing have led to a threefold increase over 10 years in the identification of monogenic diabetes in the U.K. *Diabetes Care*. 2022 (published online 21 January 2022). University of Exeter ORE record
6. Gandolfi A, Philipson LH, Naylor RN. Diagnosis and Clinical Management of Monogenic Diabetes. *Endotext*; last updated 12 August 2026. PDF
7. University of Exeter. Next generation sequencing of all known MODY genes. diabetesgenes.org. Link
8. Shields BM, et al. The development and validation of a clinical prediction model to determine the probability of MODY in patients with young-onset diabetes. *Diabetologia*. 2012;55(5):1265–1272. doi:10.1007/s00125-011-2418-8
9. ClinGen Monogenic Diabetes Variant Curation Expert Panel. ACMG/AMP variant classification specifications for HNF1A, version 3.0.0 (approved 30 June 2025). CSpec registry
10. Richards S, et al. Standards and guidelines for the interpretation of sequence variants (ACMG/AMP). *Genet Med*. 2015;17(5):405–424. doi:10.1038/gim.2015.30
11. Pearson ER, et al. Genetic cause of hyperglycaemia and response to treatment in diabetes. *Lancet*. 2003;362(9392):1275–1281. doi:10.1016/S0140-6736(03)14571-0
12. Pearson ER, et al. Switching from insulin to oral sulfonylureas in patients with diabetes due to Kir6.2 mutations. *N Engl J Med*. 2006;355(5):467–477. doi:10.1056/NEJMoa061759

Type 1 diabetes, LADA and kidney

13. Insel RA, et al. Staging presymptomatic type 1 diabetes. *Diabetes Care*. 2015;38(10):1964–1974. doi:10.2337/dc15-1419
14. Herold KC, et al. An anti-CD3 antibody, teplizumab, in relatives at risk for type 1 diabetes. *N Engl J Med*. 2019;381(7):603–613. doi:10.1056/NEJMoa1902226
15. Buzzetti R, et al. Management of latent autoimmune diabetes in adults: a consensus statement from an international expert panel. *Diabetes*. 2020;69(10):2037–2047. doi:10.2337/dbi20-0017

16. KDIGO Diabetes Work Group. KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease. *Kidney Int*. 2022;102(5S):S1–S127. doi:10.1016/j.kint.2022.06.008

National guidelines

17. Ministria e Shëndetësisë dhe Mbrojtjes Sociale (Albania). Udhërrëfyes Klinik për KSHP: Diabeti mellitus tip 2 [Primary-care clinical guideline: type 2 diabetes mellitus] (approved 18 March 2022). PDF

18. Deutsche Diabetes Gesellschaft. Praxisempfehlungen 2025 – Therapie des Typ-2-Diabetes. PDF · Contents

19. Society of Endocrinology and Metabolism of Turkey (TEMD). TEMD Diyabet Teknolojileri Kılavuzu 2025 [Diabetes Technologies Guideline 2025]. PDF

20. Society of Endocrinology and Metabolism of Turkey (TEMD). TEMD Diabetes Mellitus ve Komplikasyonlarının Tanı, Tedavi ve İzlem Kılavuzu—2018 [Guideline for the diagnosis, treatment and follow-up of diabetes mellitus and its complications] (latest edition record found; whether a newer edition exists could not be confirmed). Record

Guideline assessed

21. StrataGen International LLC; Şahiner YE. SG-END-001 Diabetes and Its Subtypes Guideline, Rev 1.2 (TR/EN/SQ/DE), 27 October 2026. The guideline's numerical features (numbers of sections, references, tables and figures) were counted from the file itself.